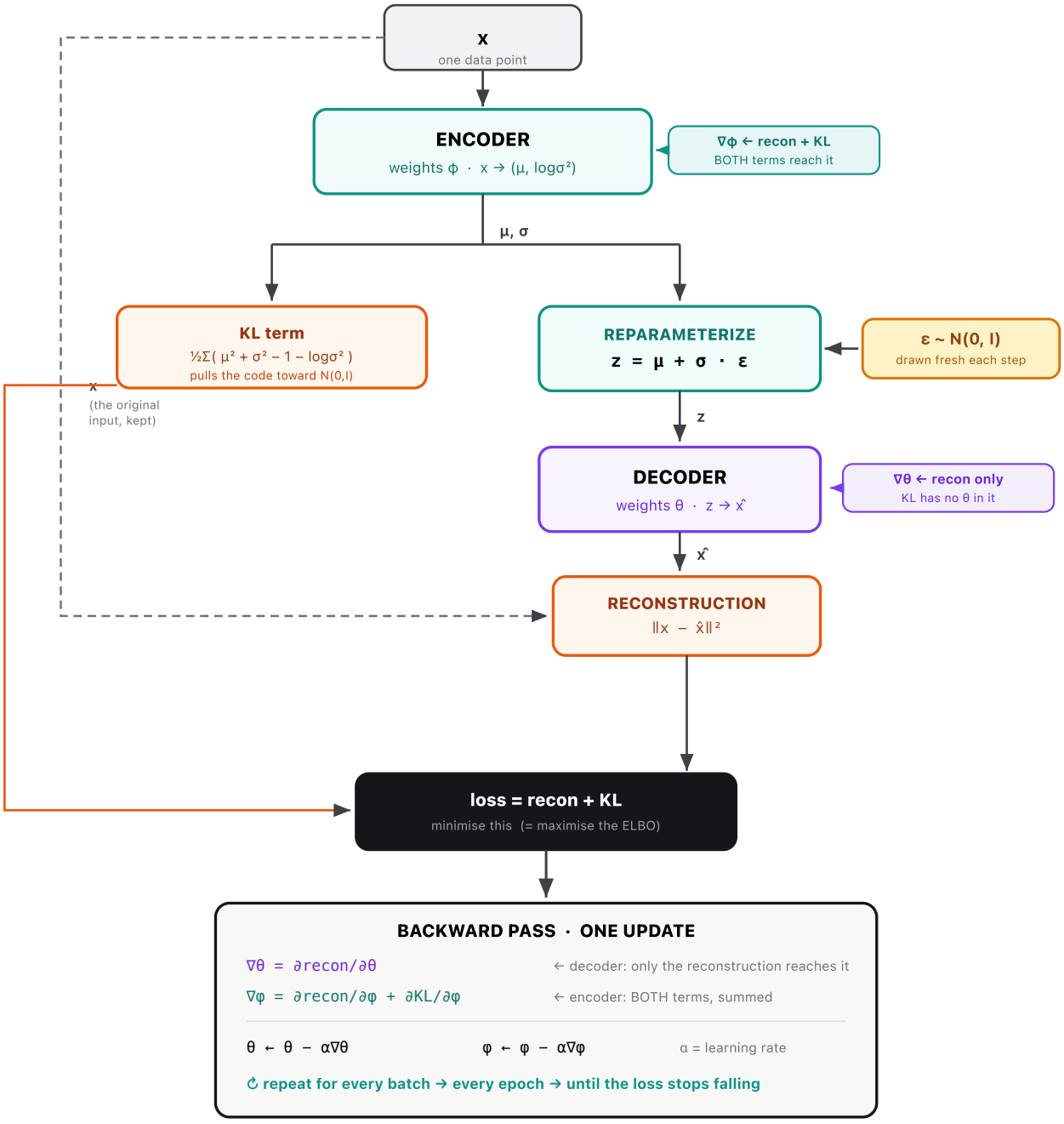
How VAE training works — end to end

One pass = one forward sweep, one loss, one backward sweep, one update. Then repeat.



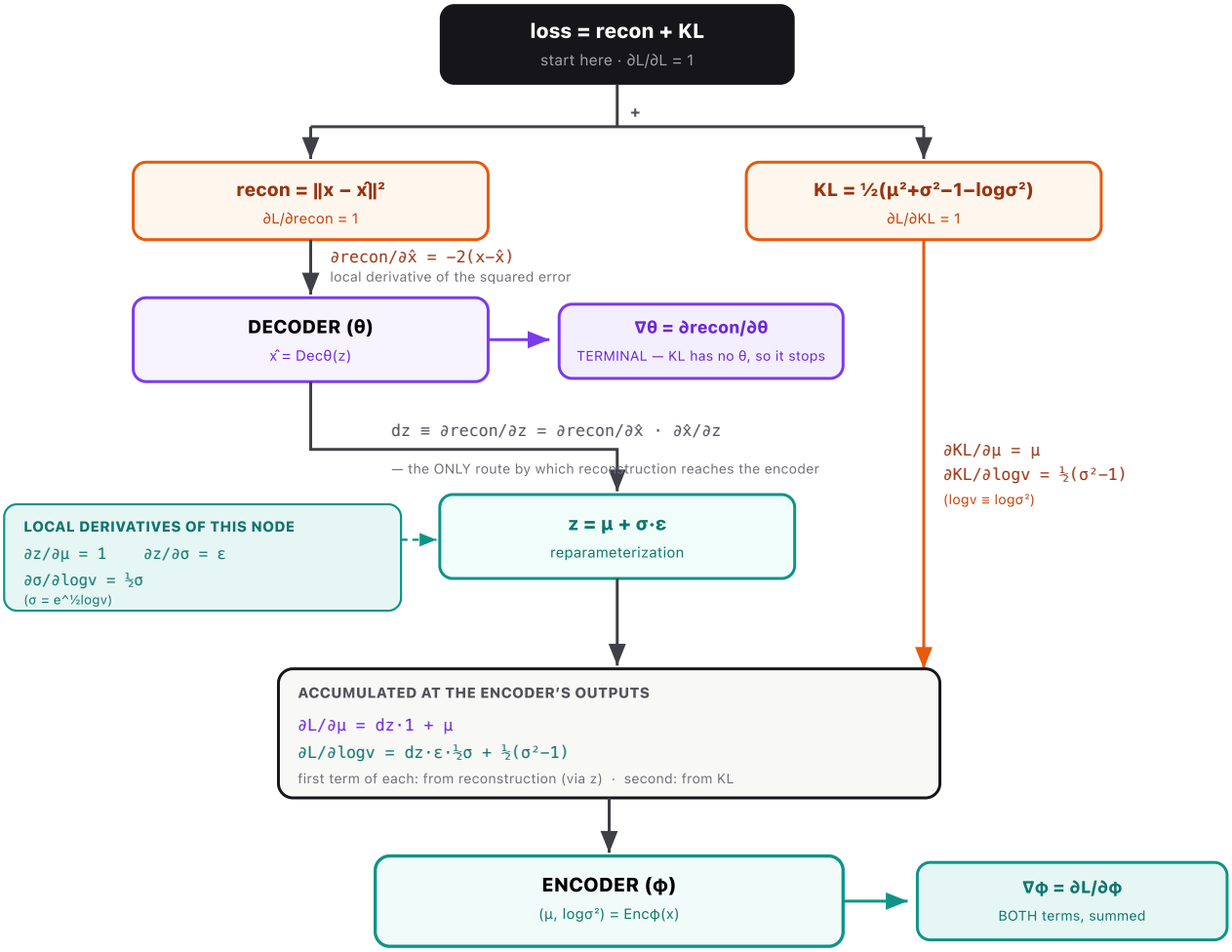
the ∇ϕ path runs **backwards through z** — that is exactly why the reparameterization block exists. Without it this arrow is severed.

forward flow is **top → bottom**; gradients flow **bottom → up**, taking **different routes** for θ and for ϕ.

Reading it: the data **x** goes into the encoder, which emits the *parameters* of a Gaussian over the code (μ and logσ²) — not a code. A fresh noise sample **ε** is combined with them to produce **z**. The decoder maps **z** to a reconstruction **x̂**. The two loss terms are computed from different inputs (**x** & **x̂** for reconstruction; **μ** & **σ** for KL) and summed. One backward pass then routes **∇θ** into the decoder alone and **∇ϕ** into the encoder from both terms.

The backward pass — every local derivative

Read **top → bottom**: the gradient is peeled backwards through the graph, one node at a time. Each edge shows its local derivative; each box shows what has accumulated.



Why the trick matters here: dz must travel from x̂ back into μ and σ. If z were drawn by an opaque `sample()`, there would be no ∂z/∂μ and no ∂z/∂σ — the arrow into the encoder would be severed and φ would never learn.

THE SAME PASS WITH NUMBERS — from the dry run (μ = 0.8, σ = 0.6, ε = −1.5, x = [1.0, −0.5], x̂ = [0.9, −0.4])

RECONSTRUCTION SIDE

```
x − x̂ = [0.10, −0.10]
∂recon/∂x̂ = −2(x−x̂) = [−0.20, +0.20]
∂z/∂μ = 1    ∂z/∂σ = ε = −1.5
∂z/∂logv = ε·½σ = −1.5 × 0.30 = −0.45
```

KL SIDE

```
∂KL/∂μ = μ = 0.80
∂KL/∂logv = ½(σ² − 1)
          = ½(0.36 − 1) = −0.32
```

PUT TOGETHER

∂L/∂μ = 1·dz + 0.80 **∂L/∂logv = −0.45·dz − 0.32**

dz = ∂recon/∂z, which needs the decoder's weights — not knowable here

The two terminals are different, and that is the whole story. θ finishes early — the reconstruction gradient reaches it and stops, because KL contains no θ. φ finishes last — it must sum a reconstruction contribution that has travelled all the way back through **z**, plus a direct KL contribution.